1 Algorithm

Algorithm 1: CNN-Adaptive Control Loop

Initialization:
  Define vehicle parameters: $m, I_z, l_f, l_r, C_{\alpha f}, C_{\alpha r}, \tau$
  Set path $P(X,Y)=0$
  Initialize vehicle state $\mathbf{x}_0 = [X_v, Y_v, \psi, v_y, r, \theta_f, \theta_r]^T$
  Pre-fill history buffer $\mathcal{W}_0$ with zeros

// Per-iteration Control Loop
for $t = 0$ **to** T_{end} **do**
  // 1. Perception & Error Extraction
  $(X^*, Y^*) \leftarrow \arg\min_{(X,Y)} (X - X_v)^2 + (Y - Y_v)^2$
    s.t. P(X,Y)=0
  $T_X \leftarrow -P_{Y^*}, \quad T_Y \leftarrow P_{X^*}$ // Unit Tangent Vector
  $T_{norm} \leftarrow \sqrt{T_X^2 + T_Y^2}$
  $\hat{T}_X \leftarrow T_X / T_{norm}, \quad \hat{T}_Y \leftarrow T_Y / T_{norm}$
  $e_y \leftarrow \operatorname{sgn}\left((X_v - X^*)\hat{T}_Y - (Y_v - Y^*)\hat{T}_X\right) \cdot \sqrt{(X_v - X^*)^2 + (Y_v - Y^*)^2}$
  $e_\psi \leftarrow \text{normalize}\left(\text{atan2}(\hat{T}_Y, \hat{T}_X) - \psi\right)$
  // 2. Neural Adaptive Model (NAM) Inference
  $\boldsymbol{\xi}_t \leftarrow [e_y, e_\psi, \omega_{f,t-1}, \omega_{r,t-1}]^T$
  $\mathcal{W}_t \leftarrow [\boldsymbol{\xi}_{t-N}, \ldots, \boldsymbol{\xi}_t]$
  $\mathbf{z} \leftarrow \text{1D-CNN}(\mathcal{W}_t)$ // Spatial-temporal feature extraction
  $\mathbf{u}_t = [\omega_{f,t}, \omega_{r,t}]^T \leftarrow \text{MLP}(\mathbf{z})$ // Adaptive control output
  // 3. Plant Dynamics Update
  $\dot{\theta}_{f,r} \leftarrow \frac{1}{\tau}(\omega_{f,r} - \theta_{f,r})$ // Actuator lag
  $\alpha_f \leftarrow \theta_f - \frac{v_y + l_f r}{v_x}, \quad \alpha_r \leftarrow \theta_r - \frac{v_y - l_r r}{v_x}$
  $F_{yf,yr} \leftarrow F_{max} \tanh\left(\frac{C_\alpha}{F_{max}} \alpha_{f,r}\right)$
  // 4. Numerical Integration
  $\dot{v}_y \leftarrow \frac{F_{yf} + F_{yr}}{m} - v_x r, \quad \dot{r} \leftarrow \frac{l_f F_{yf} - l_r F_{yr}}{I_z}$
  $\dot{X}_v \leftarrow v_x \cos\psi - v_y \sin\psi, \quad \dot{Y}_v \leftarrow v_x \sin\psi + v_y \cos\psi, \quad \dot{\psi} \leftarrow r$
  Update state $\mathbf{x}_{t+1}$